

Rainbow in a Glass

Density Tower Experiment



What Is a Rainbow?

Rainbows are made of colors that shine
when sunlight meets raindrops!

Did you know?
Rainbows have 7 colors!
Red, Orange, Yellow, Green,
Blue, Indigo, Violet



What Is Density?

Density means how heavy something is for its size.

A rock sinks because it is dense.
A leaf floats because it is less dense.



Today's Experiment

Let's make a colorful rainbow in a glass using liquids with different densities!

Each liquid is a different weight.
The heavy ones sink to the bottom.
The light ones float on top!



Materials You Need

1. Honey
2. Dish soap
3. Water + food coloring (blue)
4. Vegetable oil
5. Rubbing alcohol + food coloring (red)
6. Clear tall glass or jar
7. Spoon or dropper
8. Measuring cups



Step 1 — Pour Honey

Carefully pour honey into the glass.

Honey is the heaviest liquid. It will stay at the very bottom of your rainbow!



Step 2 — Add Dish Soap

Slowly add dish soap on top of the honey.

Pour gently so the layers don't mix! The soap is lighter than honey but heavier than water.



Step 3 — Add Colored Water

Mix water with blue food coloring and pour it on top of the dish soap.

Use a spoon to pour slowly.
This helps keep the layers
separate and neat!



Step 4 — Add Vegetable Oil

Carefully pour vegetable oil
on top of the water.

Oil is lighter than water,
so it floats on top!
Pour very slowly.



Step 5 — Add Rubbing Alcohol

Add rubbing alcohol colored with red food coloring on top of the oil.

This is the lightest liquid.
It stays right on top!
Ask an adult for help with this step.



Look at Your Rainbow!

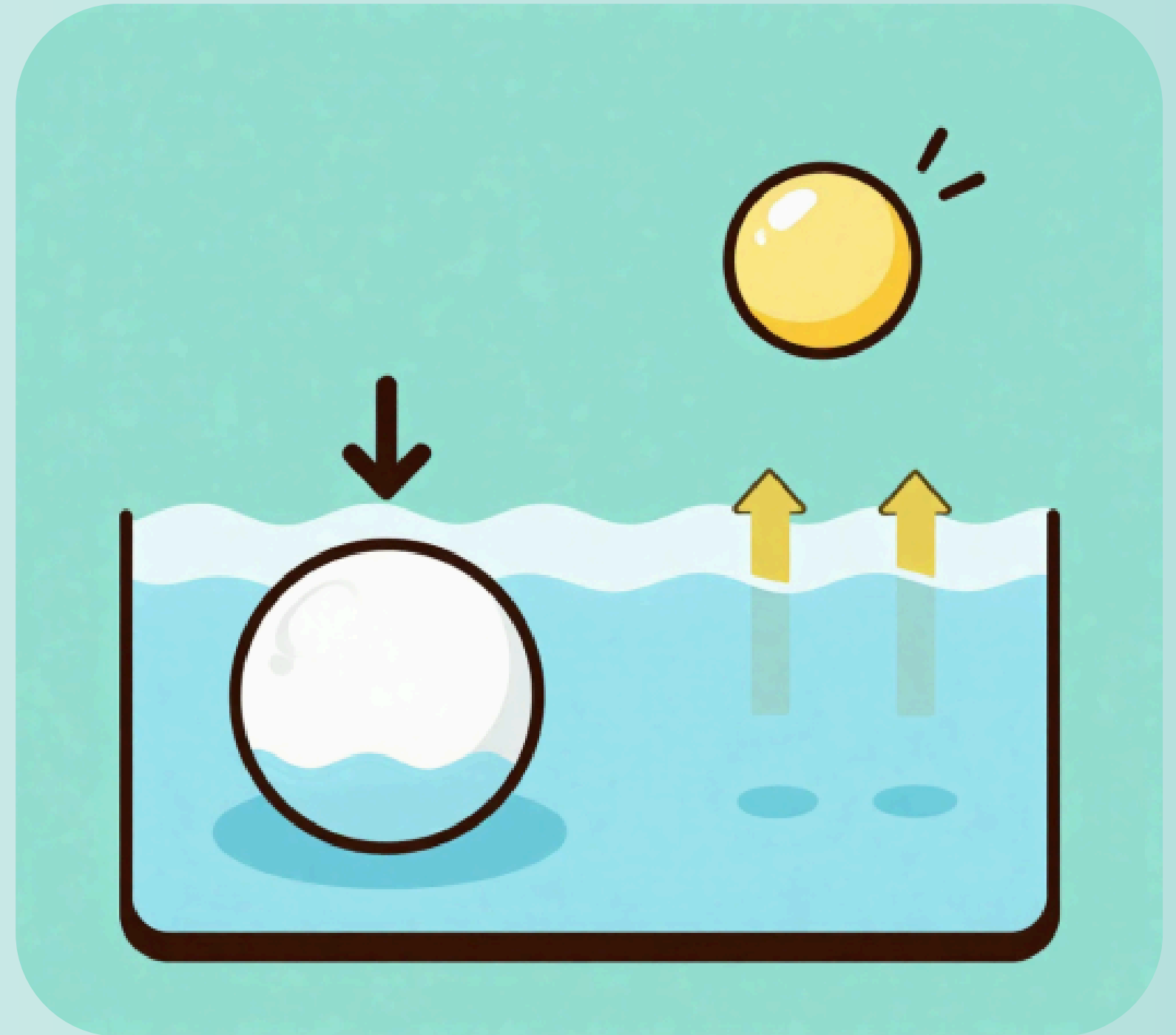
Wow! See the rainbow of colors stacked in the glass!



Why Do the Layers Stay?

Each liquid has a different density,
so they don't mix!

The heaviest stays at the bottom.
The lightest floats on top.
It's like stacking balls from
heaviest to lightest!



Fun Fact About Density

Honey is about 1.4 times
heavier than water!

That's why honey always sinks
to the bottom of your glass!



Safety First!

Do NOT drink any liquids used in the experiment

Ask an adult for help

Clean up spills right away

Wash hands after the experiment



Clean Up Time

Rinse the glass and tools with warm water.

Put away all materials.

Wipe the table clean.

Great job keeping your space tidy!



Let's Review!

What colors did you see in your rainbow?

Which liquid was the heaviest?

Why don't the liquids mix?

What safety rules should you remember?



Thank You!

Thanks for learning with us!

You are amazing scientists!

Any questions?

